1) $f(t) = t^2 e^{-t} + 3\sin 3t - 2$

$L(f(t)) = L(t^2 e^{-t}) + 3L(\sin 3t) - L(2)$

1) $L(2) = \frac{2}{s}$

2) $L(\sin 3t) = \frac{3}{s^2+9}$

3) $L(t^2 e^{-t}) = \frac{2!}{(s+1)^3} = \frac{2}{(s+1)^3}$

$$F(s) = \frac{2}{(s+1)^3} + \frac{9}{s^2+9} - \frac{2}{s}$$

2) $F(s) = \frac{s-3}{s^2-6s+13} = \frac{s-3}{(s-3)^2+4}$

$G(s) = \frac{s}{s^2+4}$

$g(t) = \cos 2t$

$$f(t) = e^{3t}\cos 2t$$

3) $\ddot{y}(t) + 2\dot{y}(t) + 5y(t) = e^{-t}$ $\quad y(0) = 0 \quad \dot{y}(0) = 1$

$$L(y(t)) = Y(s)$$

$$L(\dot{y}(t)) = sY(s) - y(0) = sY(s)$$

$$L(\ddot{y}(t)) = s(sY(s) - y(0)) - \dot{y}(0) = s^2 Y(s) - 1$$

$$L(e^{-t}) = \frac{1}{s+1}$$

$$s^2 Y(s) - 1 + 2s\,Y(s) + 5\,Y(s) = \frac{1}{s+1}$$

$$Y(s)(s^2 + 2s + 5) = \frac{s+2}{s+1}$$

$$Y(s) = \frac{s+2}{(s+1)(s^2+2s+5)} = \frac{A}{s+1} + \frac{Bs+C}{s^2+2s+5}$$

$$A = \frac{1}{4} \quad B = -\frac{1}{4} \quad C = \frac{3}{4}$$

$$Y(s) = \frac{1}{4} \cdot \frac{1}{s+1} - \frac{1}{4} \cdot \frac{s-3}{s^2+2s+5}$$

$$L^{-1}\left(\frac{1}{s+1}\right) = e^{-t}$$

$$L^{-1}\left(\frac{s-3}{s^2+2s+5}\right) = L^{-1}\left(\frac{s+1}{(s+1)^2+4}\right) - L^{-1}\left(\frac{4}{(s+1)^2+4}\right) =$$

$$= e^{-t}\cos 2t - 2e^{-t}\sin 2t$$

$$y(t) = \frac{1}{4} \cdot e^{-t} - \frac{1}{4} e^{-t}\cos 2t + \frac{1}{4} e^{-t} \cdot 2\sin 2t$$

$$y(t) = \frac{1}{4} e^{-t}(1 - \cos 2t + 2\sin 2t)$$